

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture
LAB EXPERIMENT # 05

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Score: _____ Signature of the Lab Tutor: _____ Date: _____

VLAN One-Arm Routing

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	SCORE
OBJECTIVE NO:				

Learning Objectives:

As a result of this lab section, you should achieve the following tasks:

- Establish VLAN routing and configure sub-interfaces on a single physical interface
- Establishment of a trunk interface for VLAN routing
- Configuration of sub-interfaces on a single physical interface
- Enabling of ARP messages to be broadcast between VLANs

Keywords: One-Arm Routing, Inter-VLAN Communication

Duration: 03 hours

1 Overview of lab

1.1. Introduction

This lab demonstrates the implementation of One-Arm Routing to enable communication between multiple VLANs using a single physical interface on a router. In this setup, a trunk link is established between the switch and the router, allowing VLAN-tagged traffic (IEEE 802.1Q) to pass through. The router's single interface is divided into multiple sub interfaces, each configured with an IP address corresponding to a VLAN, along with appropriate encapsulation commands. By the end of this lab, students will understand the working of Router-on-a-Stick topology, the role of trunk links in VLAN environments, and how inter-VLAN routing is achieved efficiently without multiple physical router interfaces

1.2. Inter-VLAN Communication

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Inter-VLAN communication is the process that allows devices connected to different VLANs (Virtual Local Area Networks) within the same switched network to exchange data packets. Since VLANs are separate Layer 2 broadcast domains, devices in different VLANs cannot communicate directly by default. Inter-VLAN communication overcomes this isolation by introducing Layer 3 routing functionality (either through a router or a Layer 3 switch), enabling traffic to be forwarded between these VLANs. Inter-VLAN communication provides controlled communication while maintaining VLAN boundaries.

1. Each VLAN is assigned to a unique IP subnet

- VLAN 10 → 192.168.10.0/24, VLAN 20 → 192.168.20.0/24.

2. A Layer 3 device (router or L3 switch) has interfaces or sub-interfaces assigned to each VLAN □

These interfaces act as default gateways for hosts in their VLAN.

3. When a host in VLAN 10 wants to reach VLAN 20

- The traffic goes to the default gateway (router/L3 switch).
- The router routes the packet to the correct VLAN interface.
- The packet is delivered to the destination host in VLAN 20.

1.3. One-Arm Routing

One-Arm Routing, commonly called Router-on-a-Stick (ROAS), is a method of Inter-VLAN communication where a single physical router interface is used to route traffic between multiple VLANs on a switched network. Instead of dedicating one router port per VLAN, this single interface is logically divided into multiple subinterfaces, and each sub-interface is configured to handle traffic for a specific VLAN using VLAN tagging.

- In a network with multiple VLANs, devices in different VLANs cannot communicate directly because VLANs create isolated Layer 2 broadcast domains.
- Traditional inter-VLAN routing required one router port per VLAN, which is costly and unscalable.
- One-arm routing solves this by allowing a single router interface to carry traffic for many VLANs over one trunk link, reducing hardware requirements while still providing full Layer 3 routing.

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Topology

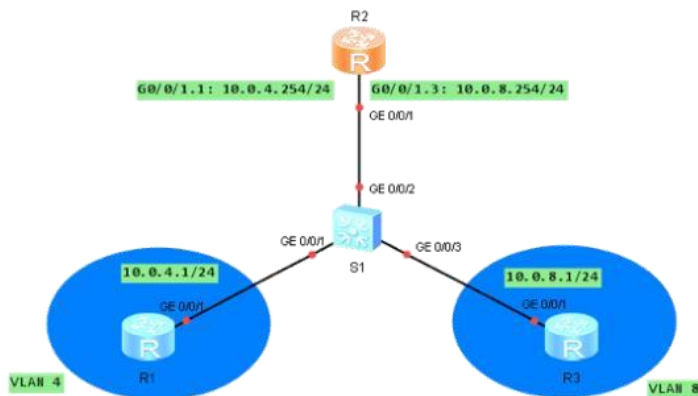
Scenario:

The implementation of VLANs in the enterprise network has resulted in groups of users being isolated from other users that are part of different subnets. As the network administrator you have been given the task to ensure that the broadcast domains are maintained whilst allowing communication between the disparate users.

Learning Objectives:

As part of this lab section, you are expected to achieve the following tasks:

1. Establishment of a trunk interface for VLAN routing.
2. Configuration of sub-interfaces on a single physical interface.
3. Enabling of ARP messages to be broadcast between VLANs.



Configure IP address and default route

```
<Huawei>system-view
[Huawei]sysname R1
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip add 10.0.4.1 24
[R1-GigabitEthernet0/0/1]q
[R1]ip route-static 0.0.0.0 0.0.0.0 10.0.4.254

<Huawei>system-view
[Huawei]sysname R3
[R3]int g0/0/1
[R3-GigabitEthernet0/0/1]ip add 10.0.8.1 24
[R3-GigabitEthernet0/0/1]q
[R3]ip route-static 0.0.0.0 0.0.0.0 10.0.8.254
```

Test connectivity between R1 and R3

```
<R1>ping -c 50 10.0.8.1
```

Establish two VLANs

```
<Huawei>system-view
[Huawei]sysname S1
[S1]vlan batch 4 8
[S1]int g0/0/1
[S1-GigabitEthernet0/0/1]port link-type access
[S1-GigabitEthernet0/0/1]port default vlan 4
[S1-GigabitEthernet0/0/1]q
[S1]int g0/0/3
[S1-GigabitEthernet0/0/3]port link-type access
[S1-GigabitEthernet0/0/3]port default vlan 8
[S1-GigabitEthernet0/0/3]q
[S1]int g0/0/2
[S1-GigabitEthernet0/0/2]port link-type trunk
[S1-GigabitEthernet0/0/2]port trunk allow-pass vlan 4 8
[S1-GigabitEthernet0/0/2]q
```

Configure VLAN routing through sub-interface of R2

```
<Huawei>system-view
[Huawei]sysname R2
[R2]int g0/0/1.1
[R2-GigabitEthernet0/0/1.1]ip add 10.0.4.254 24
[R2-GigabitEthernet0/0/1.1]dot1q termination vid 4
[R2-GigabitEthernet0/0/1.1]arp broadcast enable
[R2-GigabitEthernet0/0/1.1]q
[R2]int g0/0/1.3
[R2-GigabitEthernet0/0/1.3]ip add 10.0.8.254 24
[R2-GigabitEthernet0/0/1.3]dot1q termination vid 8
[R2-GigabitEthernet0/0/1.3]arp broadcast enable
[R2-GigabitEthernet0/0/1.3]q
[R2]display ip routing-table
```

- By default, arp is disabled on sub-interface
- IEEE 802.1Q encapsulation is the networking standard that supports VLANs.
- The command dot1q termination is used to forward VLAN packets via layer 3 routing

Scenario

The implementation of VLANs in the enterprise network has resulted in groups of users being isolated from other users that are part of different subnets. As the network administrator you have been given the task of ensuring that the broadcast domains are maintained whilst allowing communication between the disparate users.

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Step 1 Preparing the environment

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname R1
[R1] interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1] ip address 10.0.4.1 24
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname R3
[R3] interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1] ip address 10.0.8.1 24 <Huawei>
system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname S1
[S1]
```

Step 2 Establishing two VLANs

Create VLANs 4 and 8 on S1.

```
[S1] vlan batch 4 8
Info: This operation may take a few seconds. Please wait for a moment...done.
[S1] interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1] port link-type access
[S1-GigabitEthernet0/0/1] port default vlan 4
[S1-GigabitEthernet0/0/1] quit
[S1] interface GigabitEthernet0/0/3
[S1-GigabitEthernet0/0/3] port link-type access
[S1-GigabitEthernet0/0/3] port default vlan 8
[S1-GigabitEthernet0/0/3] quit
```

Trunk link between the switch and the router must be established to support traffic for multiple VLANs.

Set interface **Gigabit Ethernet 0/0/2** as a trunk link to pass **VLANs 4 and 8** on S1

```
[S1] interface GigabitEthernet0/0/2
[S1-GigabitEthernet0/0/2] port link-type trunk
[S1-GigabitEthernet0/0/2] port trunk allow-pass vlan 4 8
```

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Step 3 Configure VLAN routing through the sub-interface of R2

Configure sub-interfaces **Gigabit Ethernet 0/0/1.1** and **Gigabit Ethernet 0/0/1.3**, to act as the gateway of **VLAN 4**, and act as the gateway of **VLAN 8**.

The command "**dot1q termination**" is enabled under **sub-interface** to send and receive **VLAN tag frames**.

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname R2
[R2] interface GigabitEthernet0/0/1.1
[R2-GigabitEthernet0/0/1.1] ip address 10.0.4.254 24
[R2-GigabitEthernet0/0/1.1] dot1q termination vid 4
[R2-GigabitEthernet0/0/1.1] arp broadcast enable
[R2-GigabitEthernet0/0/1.1] quit
[R2] interface GigabitEthernet0/0/1.3
[R2-GigabitEthernet0/0/1.3] ip address 10.0.8.254 24
[R2-GigabitEthernet0/0/1.3] dot1q termination vid 8
[R2-GigabitEthernet0/0/1.3] arp broadcast enable
```

If **ARP broadcast** remains disabled on the sub-interface, the router will directly discard packets.

Configure a default route on **R1** and **R3**.

```
[R1] ip route-static 0.0.0.0 0.0.0.0 10.0.4.254 [R3]
ip route-static 0.0.0.0 0.0.0.0 10.0.8.254
```

Test connectivity between R1 and R3 again.

```
<R1> ping 10.0.8.1
PING 10.0.8.1: 56 data bytes, press CTRL_C to break
Reply from 10.0.8.1: bytes=56 Sequence=1 ttl=254 time=10 ms
Reply from 10.0.8.1: bytes=56 Sequence=2 ttl=254 time=1 ms
Reply from 10.0.8.1: bytes=56 Sequence=3 ttl=254 time=1 ms
Reply from 10.0.8.1: bytes=56 Sequence=4 ttl=254 time=10 ms
Reply from 10.0.8.1: bytes=56 Sequence=5 ttl=254 time=1 ms

--- 10.0.8.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
```

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round-trip min/avg/max = 1/4/10 ms

Observe the routing table on R2.

```
[R2] display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 10 Routes : 10

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.4.0/24	Direct	0	0	D	10.0.4.254	GigabitEthernet0/0/1.1
10.0.4.254/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1.1
10.0.4.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1.1
10.0.8.0/24	Direct	0	0	D	10.0.8.254	GigabitEthernet0/0/1.3
10.0.8.254/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1.3
10.0.8.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1.3
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0 127.0.0.1/32
Direct 0 0 D			127.0.0.1		InLoopBack0	
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

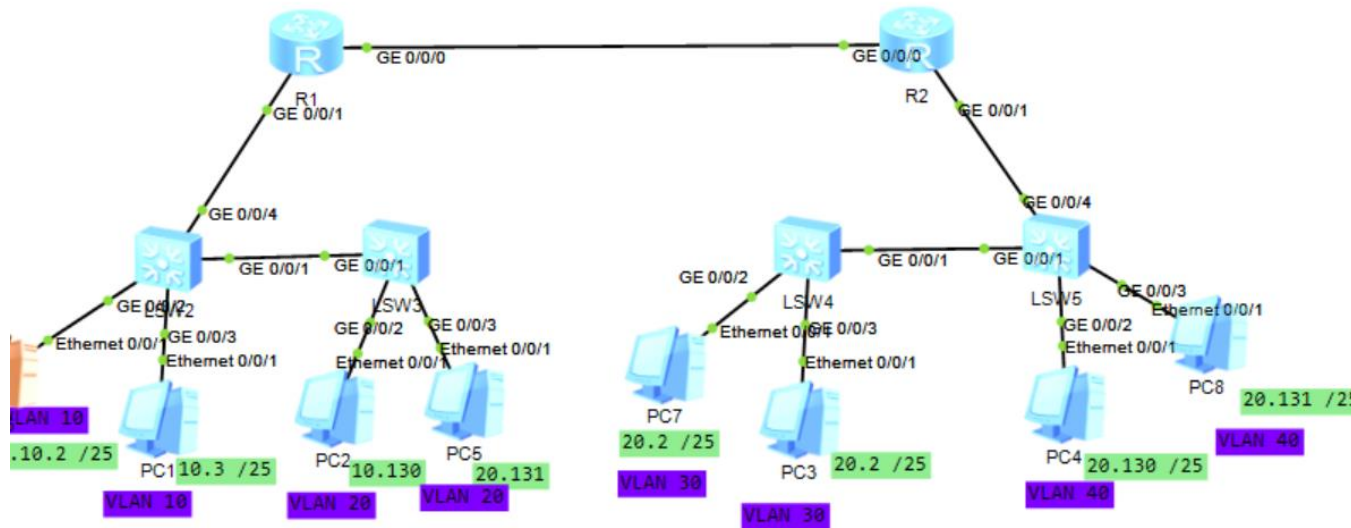
Tasks

What is the purpose of dot1q termination vid 4?

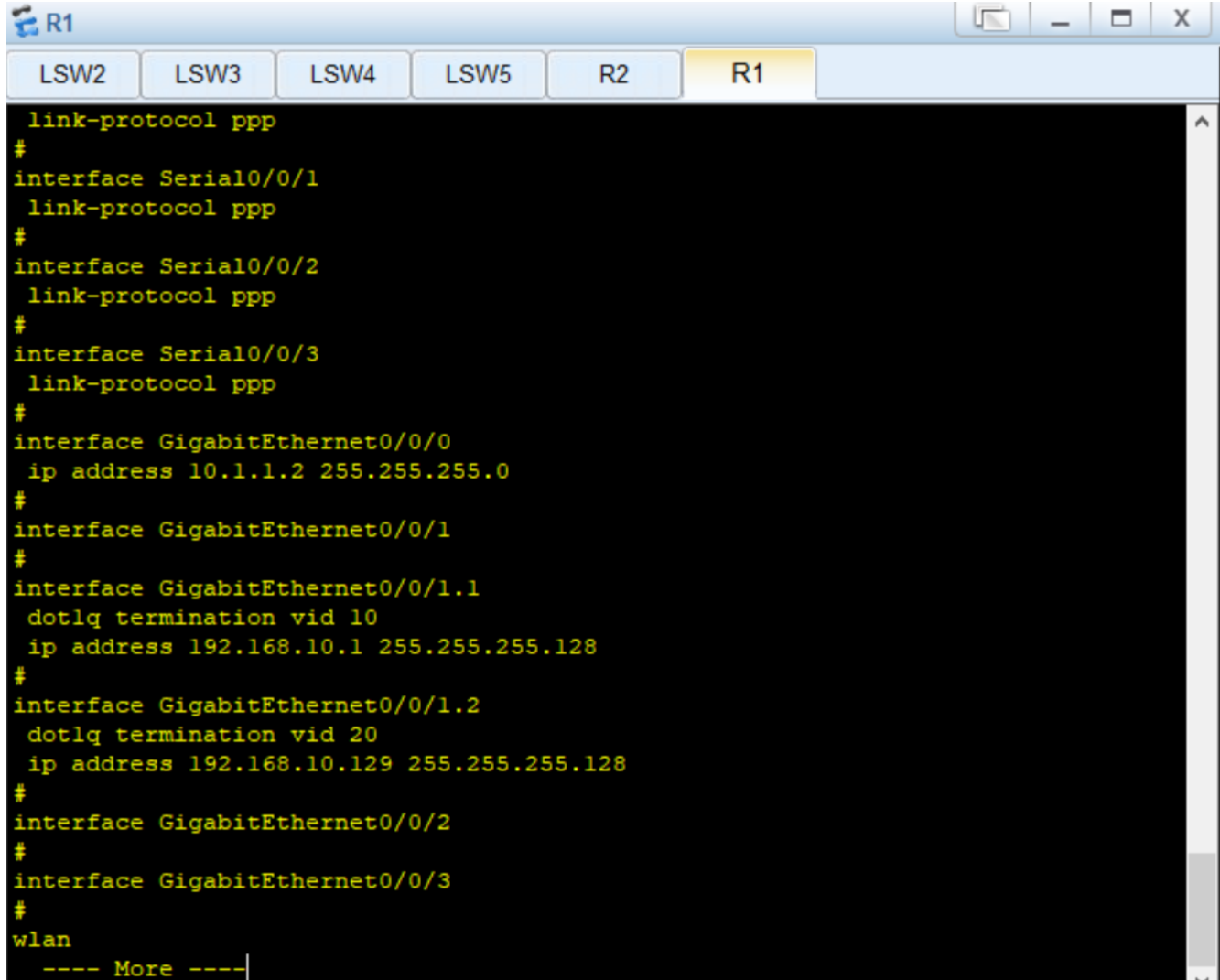
Try a different topology and demonstrate your learning

Upload the `.topo` `.efz` `.cfg` files

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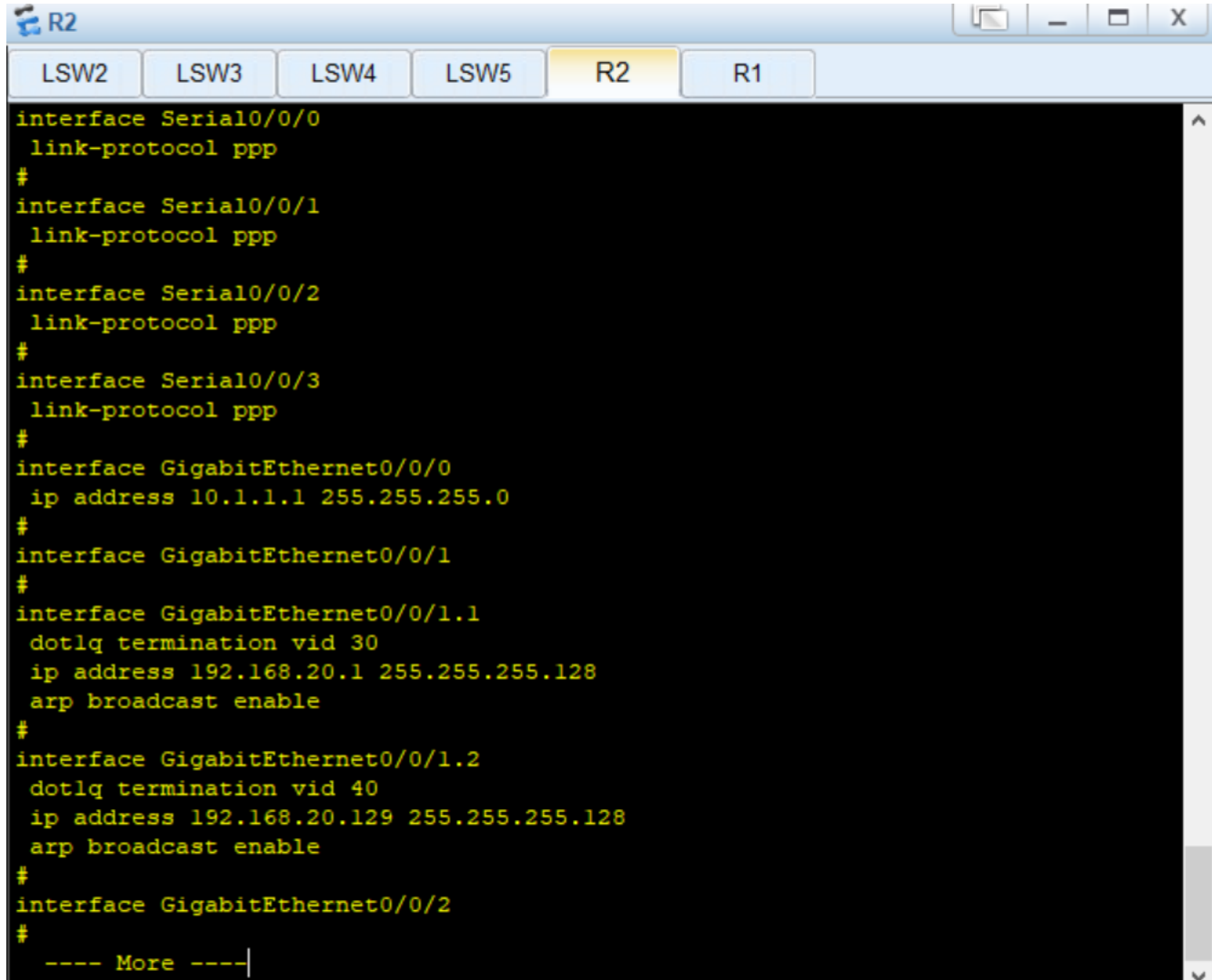


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```
link-protocol ppp
#
interface Serial0/0/1
  link-protocol ppp
#
interface Serial0/0/2
  link-protocol ppp
#
interface Serial0/0/3
  link-protocol ppp
#
interface GigabitEthernet0/0/0
  ip address 10.1.1.2 255.255.255.0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/1.1
  dot1q termination vid 10
  ip address 192.168.10.1 255.255.255.128
#
interface GigabitEthernet0/0/1.2
  dot1q termination vid 20
  ip address 192.168.10.129 255.255.255.128
#
interface GigabitEthernet0/0/2
#
interface GigabitEthernet0/0/3
#
wlan
---- More ----
```


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```
interface Serial0/0/0
  link-protocol ppp
#
interface Serial0/0/1
  link-protocol ppp
#
interface Serial0/0/2
  link-protocol ppp
#
interface Serial0/0/3
  link-protocol ppp
#
interface GigabitEthernet0/0/0
  ip address 10.1.1.1 255.255.255.0
#
interface GigabitEthernet0/0/1
#
interface GigabitEthernet0/0/1.1
  dot1q termination vid 30
  ip address 192.168.20.1 255.255.255.128
  arp broadcast enable
#
interface GigabitEthernet0/0/1.2
  dot1q termination vid 40
  ip address 192.168.20.129 255.255.255.128
  arp broadcast enable
#
interface GigabitEthernet0/0/2
#
---- More ----|
```